Linear System Fall 2026

Homework Five

Due: Friday, Oct 31

Exercise 1 Suppose A is a 3×3 matrix and

$$\det(sI - A) = s^3 + 2s^2 + s + 1.$$

(a) Express A^3 in terms of I, A, and A^2.

(b) Express A^5 in terms of I, A, and A^2.

(c) Express A^{-1} in terms of I, A, and A^2.

Exercise 2 Without doing any matrix multiplications, compute A^4 for

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 564 & 0 & 0 & 0 \end{bmatrix}.$$

Justify your answer.

Exercise 3 Consider the differential equation

$$\dot{x} = Ax,$$

where A is a square matrix. Show that if A has $j2\pi$ as an eigenvalue, then there is a nonzero initial state x_0 such that the solution x satisfies

$$x(1) = x(0) = x_0.$$

Exercise 4 Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

by:

(a) use Jordan form using eigenvalues and eigenvectors of A;

(b) using the Laplace transform.

Exercise 5 Compute e^{At} for

$$A = \begin{bmatrix} 3 & -5 \\ -5 & 3 \end{bmatrix}.$$

by:

(a) use Jordan form using eigenvalues and eigenvectors of A;

(b) using the Laplace transform.

Exercise 6 Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Exercise 7 Obtain (by hand) the state response and output response $x(\cdot)$ of the following system due to a unit impulse input and the zero initial conditions. Determine whether the state response contains all the system modes.

$$\dot{x}_1 = -5x_1 + 2x_2 + u$$

$$\dot{x}_2 = -12x_1 + 5x_2 + u$$

$$y = 2x_1 + 3x_2$$